

Lie Theory: Chapter 3 Homework

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1. Let L be a non-abelian 3-dimensional Lie algebra with basis $\{f, g, z\}$ and $[f, g] := z$. Assume $z \in Z(L)$ and L' is one-dimensional. Prove that $L' = Z(L)$.

Proof. First, we realize that a general element of L' is

$$a[f, g] + b[f, z] + c[g, z] = az$$

because $[f, g] = z$ and $[f, z] = [g, z] = 0$ since z is in the center. Since we are given that L' is one-dimensional, it must be spanned by z . Thus when we bracket a general element of L' with a general element of L :

$$[\alpha z, af + bg + cz] = \alpha[z, af + bg + cz] = 0$$

since $z \in Z(L)$. Thus $L' \subset Z(L)$.

Now we examine a general element of the center, $af + bg + cz$, bracketed with a general element of L :

$$\begin{aligned} 0 &= [af + bg + cz, \alpha f + \beta g + \gamma z] \\ &= a\beta[f, g] + a\gamma[f, z] + b\alpha[g, f] + b\gamma[g, z] + c\alpha[z, f] + c\beta[z, g] \\ &= a\beta z - b\alpha z \\ &= (a\beta - b\alpha)z. \end{aligned}$$

Since $z \neq 0$, we have $a\beta = b\alpha$ for every choice of α and β . In particular, if $\alpha = 0$ and $\beta = 1$, we get that $a = 0$, and if $\alpha = 1$ but $\beta = 0$, we have $0 = b$. Thus $af + bg + cz = cz$, where c is truly free to vary over F , so $Z(L)$ is one-dimensional. We already showed that $L' \subset Z(L)$, so since they have the same dimension, $L' = Z(L)$. $\square$

2. Check that the strictly upper triangular matrices $\{e_{12}, e_{23}, e_{13}\}$ satisfy the following properties.

(a) $[e_{12}, e_{23}] = e_{13}$.

(b) e_{13} is in the center of the Lie algebra of strictly upper triangular matrices.

This shows that the set $\{e_{12}, e_{23}, e_{13}\}$ satisfies the construction described in Section 3.2.1 with $z = e_{13}$ and f and g equal to e_{12} and e_{23} .

Proof.

(a)

$$\begin{aligned}
[e_{12}, e_{23}] &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= e_{13}.
\end{aligned}$$

(b) We bracket e_{13} with a general element of $b(3, F)$:

$$\begin{aligned}
\left[\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix} \right] &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}
\end{aligned}$$

so $e_{13} \in Z(n(3, F))$ as desired.

□

3. Let V be a vector space and let φ be an endomorphism of V . Let L have underlying vector space $V \oplus \text{Span}\{x\}$. Show that if we define the Lie bracket on L by $[y, z] = 0$ and $[x, y] = \varphi(y)$ for $y, z \in V$, then L is a Lie algebra and $\dim(L') = \text{rank}(\varphi)$.

Proof. Given any $y, z \in V$ and $a, b \in F$, we represent a general bracket in L as follows:

$$\begin{aligned}
[(y, ax), (z, bx)] &= [(y, 0) + (0, ax), (z, 0) + (0, bx)] \\
&= [(y, 0), (z, 0)] + [(y, 0), (0, bx)] + [(0, ax), (z, 0)] + [(0, ax), (0, bx)] \\
&= 0 + [(y, 0), b(0, x)] + [a(0, x), (z, 0)] + [a(0, x), b(0, x)] \\
&= b[(y, 0), (0, x)] + a[(0, x), (z, 0)] + ab[(0, x), (0, x)] \\
&= -b\varphi(y) + a\varphi(z) \\
&= a\varphi(z) - b\varphi(y) \\
&= (a\varphi(z) - b\varphi(y), 0) \\
&= (\varphi(az - by), 0),
\end{aligned}$$

where we will frequently use the abuse-of-notation third-to-last line for brevity. Now we can easily check the conditions for a bracket (bilinearity is already assumed). First,

$$[(y, ax), (y, ax)] = a\varphi(y) - a\varphi(y) = 0$$

as desired, and

$$\begin{aligned} & [(w, ax), [(y, bx), (z, cx)]] + [(y, bx), [(z, cx), (w, ax)]] + [(z, cx), [(w, ax), (y, bx)]] \\ &= [(w, ax), b\varphi(z) - c\varphi(y)] + [(y, bx), c\varphi(w) - a\varphi(z)] + [(z, cx), a\varphi(y) - b\varphi(w)] \\ &= [(w, ax), (b\varphi(z) - c\varphi(y), 0)] + [(y, bx), (c\varphi(w) - a\varphi(z), 0)] + [(z, cx), (a\varphi(y) - b\varphi(w), 0)] \\ &= a\varphi(b\varphi(z) - c\varphi(y)) + b\varphi(c\varphi(w) - a\varphi(z)) + c\varphi(a\varphi(y) - b\varphi(w)) \\ &= ab\varphi^2(z) - ac\varphi^2(y) + bc\varphi^2(w) - ab\varphi^2(z) + ac\varphi^2(y) - bc\varphi^2(w) \\ &= 0, \end{aligned}$$

where the final equality comes from canceling like terms. Thus L is a Lie algebra.

Returning to our general formulation of the bracket in L where

$$[(y, ax), (z, bx)] = (\varphi(az - by), 0),$$

we see that if $z \in V$, then $[(0, x), (z, 0)] = (\varphi(z), 0)$, so $\text{im } \varphi \times \{0\} \subset L'$, and we have $\dim(\text{im } \varphi \times \{0\}) = \dim \text{im } \varphi = \text{rank}(\varphi) \leq \dim(L')$. On the other hand, we have shown that the brackets of L have the form $(\varphi(v), 0)$ for some $v \in V$, so any element of L' is a linear combination of such elements, namely something like this:

$$(\varphi(v_1 + \cdots + v_n), 0).$$

Thus $L' \subset \text{im } \varphi \times \{0\}$,

$$\dim(L') \leq \dim(\text{im } \varphi \times \{0\}) = \dim \text{im } \varphi = \text{rank} \varphi$$

Putting the two inequalities together, we get that $\dim(L') = \text{rank}(\varphi)$ as desired. $\square$

4. Suppose L is a vector space with basis $\{x, y\}$ and that a bilinear operation on L is defined such that $[u, u] = 0$ for all $u \in L$. Show that the Jacobi identity holds and hence L is a Lie algebra.

Proof. Since $[u, u] = 0$ for all $u \in L$, we have the familiar fact that $[u, v] = -[v, u]$. We also still have the core fact about structure constants, namely that the proposed bracket depends entirely on $[x, y]$. Let α, β be the constants satisfying $[x, y] = \alpha x + \beta y$. Thus we can go ahead and compute that $[x, [x, y]] = [x, \alpha x + \beta y] = \beta[x, y]$, and similarly $[y, [x, y]] = [y, \alpha x + \beta y] =$

$-\alpha[x, y]$. We have

$$\begin{aligned}
& [a_1x + b_1y, [a_2x + b_2y, a_3x + b_3y]] \\
& + [a_2x + b_2y, [a_3x + b_3y, a_1x + b_1y]] \\
& + [a_3x + b_3y, [a_1x + b_1y, a_2x + b_2y]] \\
= & [a_1x + b_1y, (a_2b_3 - a_3b_2)[x, y]] \\
& + [a_2x + b_2y, (a_3b_1 - a_1b_3)[x, y]] \\
& + [a_3x + b_3y, (a_1b_2 - a_2b_1)[x, y]] \\
= & [a_1x, (a_2b_3 - a_3b_2)[x, y]] + [b_1y, (a_2b_3 - a_3b_2)[x, y]] \\
& + [a_2x, (a_3b_1 - a_1b_3)[x, y]] + [b_2y, (a_3b_1 - a_1b_3)[x, y]] \\
& + [a_3x, (a_1b_2 - a_2b_1)[x, y]] + [b_3y, (a_1b_2 - a_2b_1)[x, y]] \\
= & a_1(a_2b_3 - a_3b_2)[x, [x, y]] + b_1(a_2b_3 - a_3b_2)[y, [x, y]] \\
& + a_2(a_3b_1 - a_1b_3)[x, [x, y]] + b_2(a_3b_1 - a_1b_3)[y, [x, y]] \\
& + a_3(a_1b_2 - a_2b_1)[x, [x, y]] + b_3(a_1b_2 - a_2b_1)[y, [x, y]] \\
= & \alpha a_1(a_2b_3 - a_3b_2)[x, y] - \beta b_1(a_2b_3 - a_3b_2)[x, y] \\
& + \alpha a_2(a_3b_1 - a_1b_3)[x, y] - \beta b_2(a_3b_1 - a_1b_3)[x, y] \\
& + \alpha a_3(a_1b_2 - a_2b_1)[x, y] - \beta b_3(a_1b_2 - a_2b_1)[x, y] \\
= & (\alpha a_1a_2b_3 - \alpha a_1a_3b_2 - \beta b_1a_2b_3 + \beta b_1a_3b_2 \\
& + \alpha a_2a_3b_1 - \alpha a_2a_1b_3 - \beta b_2a_3b_1 + \beta b_2a_1b_3 \\
& + \alpha a_3a_1b_2 - \alpha a_3a_2b_1 - \beta b_3a_1b_2 + \beta b_3a_2b_1)[x, y] \\
= & 0[x, y] \\
= & 0,
\end{aligned}$$

where second-to-last equality comes from canceling out like terms. Thus the Jacobi identity is satisfied. $\square$

5. Suppose that I is an ideal of a Lie algebra L and that there is a subalgebra S of L such that $L = S \oplus I$. Show that the map $\theta : S \rightarrow \mathfrak{gl}(I)$ defined by $\theta(s)(x) = [s, x]$ is a Lie algebra homomorphism from S into $\text{Der}(I)$.

We say that L is a *semidirect product* of I by S .

Proof. First we show that $\theta(s)$ is a derivation on I for all $s \in S$ by first noting that indeed $[s, x] \in I$ whenever $x \in I$ since I is an ideal, then showing that it is linear by

$$\begin{aligned}
\theta(s)(ax + by) &= [s, ax + by] \\
&= a[s, x] + b[s, y] \\
&= a\theta(s)(x) + b\theta(s)(y)
\end{aligned}$$

as desired, then showing that it satisfies the Leibniz rule:

$$\begin{aligned}
\theta(s)([x, y]) &= [s, [x, y]] \\
&= [x, [s, y]] + [[s, x], y] \\
&= [x, \theta(s)(y)] + [\theta(s)(x), y].
\end{aligned}$$

Now we show that θ is linear:

$$\begin{aligned}\theta(as_1 + bs_2)(x) &= [as_1 + bs_2, x] \\ &= a[s_1, x] + b[s_2, x] \\ &= a\theta(s_1)(x) + b\theta(s_2)(x),\end{aligned}$$

and finally we show that θ preserves the bracket:

$$\begin{aligned}\theta([s_1, s_2])(x) &= [[s_1, s_2], x] \\ &= -[x, [s_1, s_2]] \\ &= [s_1, [s_2, x]] + [s_2, [x, s_1]] \\ &= [s_1, [s_2, x]] - [s_2, [s_1, x]] \\ &= \theta(s_1)([s_2, x]) - \theta(s_2)([s_1, x]) \\ &= \theta(s_1)(\theta(s_2)(x)) - \theta(s_2)(\theta(s_1)(x)) \\ &= (\theta(s_1) \circ \theta(s_2))(x) - (\theta(s_2) \circ \theta(s_1))(x) \\ &= (\theta(s_1) \circ \theta(s_2) - \theta(s_2) \circ \theta(s_1))(x) \\ &= [\theta(s_1), \theta(s_2)](x).\end{aligned}$$

Thus θ is a Lie algebra homomorphism $S \rightarrow \text{Der}(I)$ as desired. $\square$

6. Show conversely that given Lie algebras S and I and a Lie algebra homomorphism $\theta : S \rightarrow \text{Der}(I)$, the vector space $S \oplus I$ may be made into a Lie algebra by defining

$$[(s_1, x_1), (s_2, x_2)] = ([s_1, s_2], [x_1, x_2] + \theta(s_1)x_2 - \theta(s_2)x_1)$$

for $s_1, s_2 \in S$, and $x_1, x_2 \in I$, and that this Lie algebra is a semidirect product of I by S . (The direct sum construction introduced in Exercise 2.6 is the special case where $\theta(s) = 0$ for all $s \in S$.)

Proof. First we check that the proposed bracket is bilinear:

$$\begin{aligned}& [a(s_1, x_1) + b(s_2, x_2), (s_3, x_3)] \\ &= [(as_1 + bs_2, ax_1 + bx_2), (s_3, x_3)] \\ &= ([as_1 + bs_2, s_3], [ax_1 + bx_2, x_3] + \theta(as_1 + bs_2)x_3 - b\theta(s_3)x_2) \\ &= (a[s_1, s_3] + b[s_2, s_3], a[x_1, x_3] + b[x_2, x_3] + a\theta(s_1)x_3 + b\theta(s_2)x_3 - b\theta(s_3)x_2) \\ &= (a[s_1, s_3], a[x_1, x_3] + a\theta(s_1)x_3 - a\theta(s_3)x_1) + (b[s_2, s_3], b[x_2, x_3] + b\theta(s_2)x_3 - b\theta(s_3)x_2) \\ &= a([s_1, s_3], [x_1, x_3] + \theta(s_1)x_3 - \theta(s_3)x_1) + b([s_2, s_3], [x_2, x_3] + \theta(s_2)x_3 - \theta(s_3)x_2) \\ &= a[(s_1, x_1), (s_3, x_3)] + b[(s_2, x_2), (s_3, x_3)].\end{aligned}$$

The proof follows similarly for the second slot.

Now we check the two bracket conditions. First,

$$\begin{aligned}[(s, x), (s, x)] &= ([s, s], [x, x] + \theta(s)x - \theta(s)x) \\ &= ([s, s], [x, x]) \\ &= (0, 0).\end{aligned}$$

Now to check the Jacobi identity, we have

$$\begin{aligned}
& [(s_1, x_1), [(s_2, x_2), (s_3, x_3)]] \\
& + [(s_2, x_2), [(s_3, x_3), (s_1, x_1)]] \\
& + [(s_3, x_3), [(s_1, x_1), (s_2, x_2)]] \\
= & [(s_1, x_1), ([s_2, s_3], [x_2, x_3] + \theta(s_2)x_3 - \theta(s_3)x_2)] \\
& + [(s_2, x_2), ([s_3, s_1], [x_3, x_1] + \theta(s_3)x_1 - \theta(s_1)x_3)] \\
& + [(s_3, x_3), ([s_1, s_2], [x_1, x_2] + \theta(s_1)x_2 - \theta(s_2)x_1)] \\
= & ([s_1, [s_2, s_3]], [x_1, [x_2, x_3] + \theta(s_2)x_3 - \theta(s_3)x_2]) \\
& + [(s_2, x_2), ([s_3, s_1], [x_3, x_1] + \theta(s_3)x_1 - \theta(s_1)x_3)] \\
& + [(s_3, x_3), ([s_1, s_2], [x_1, x_2] + \theta(s_1)x_2 - \theta(s_2)x_1)] \\
= & \dots \\
= & 0
\end{aligned}$$

This was getting so tedious and time-consuming that I didn't have time to get back to it throughout the week. □